

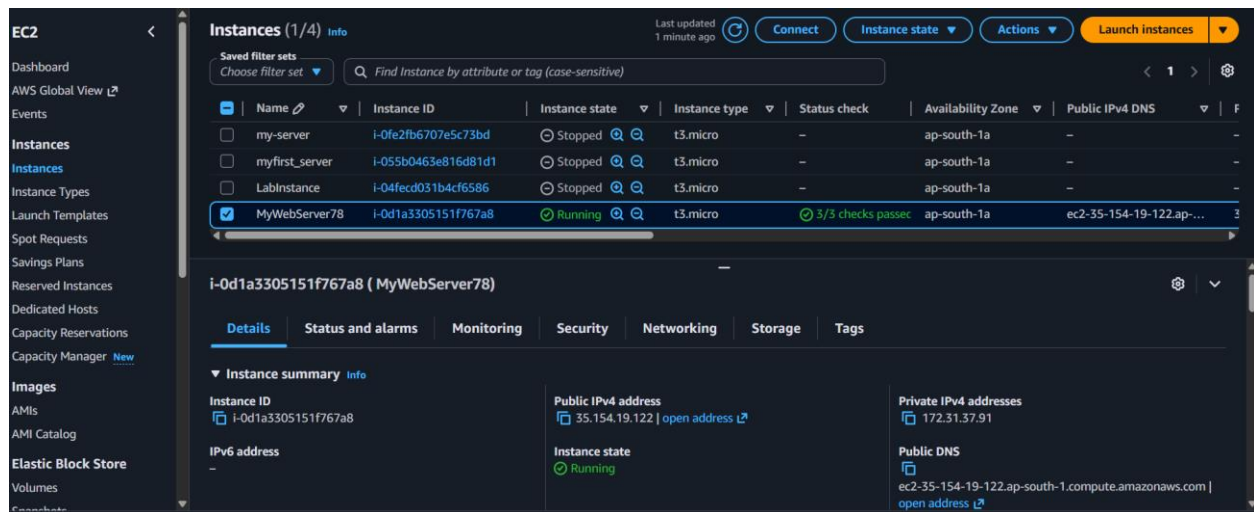
Project

EC2-based Web Server

In this project, we will deploy a web server using Amazon EC2. You will launch an EC2 instance, install either Apache or Nginx, and configure it to serve a basic website.

Step 1: Launch an EC2 Instance

1. Go to the AWS Management Console → EC2.
2. Click Launch Instance.
3. Configure the instance:
 - Name: MyWebServer
 - AMI: Choose Amazon Linux 2 or Ubuntu
 - Instance Type: t2.micro (Free Tier eligible)
 - Key Pair: Create a new key pair or use an existing one.
 - Security Group: Allow SSH (port 22) and HTTP (port 80).
4. Click Launch.



Step 2: Connect to the EC2 Instance

Open a terminal and run:

```
ssh -i your-key.pem ec2-user@your-instance-public-ip
```

Step 3: Install Apache

For Apache (httpd)

Update the package repository: `sudo yum update -y` # Amazon Linux

Install Apache:

`sudo yum install httpd -y` # Amazon Linux

Start and enable Apache:

`sudo systemctl start httpd` # Amazon Linux

`sudo systemctl enable httpd` # Amazon Linux

Verify installation:

`sudo systemctl status httpd` # Amazon Linux

```
ast metadata expiration check: 0:00:26 ago on Sun Jun 21 06:44:31 2026.
Package httpd-2.4.67-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
ec2-user@ip-172-31-37-91 ~]$ sudo systemctl start httpd
ec2-user@ip-172-31-37-91 ~]$ sudo systemctl enable httpd
ec2-user@ip-172-31-37-91 ~]$ sudo systemctl status httpd
httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-06-17 08:18:05 UTC; 3 days ago
     Docs: man:httpd.service(8)
   Main PID: 63637 (httpd)
  Status: "Total requests: 541; Idle/Busy workers 100/0; Requests/sec: 0.00159; Bytes served/sec: 1 B/sec"
    Tasks: 177 (limit: 1014)
   Memory: 15.3M
      CPU: 5min 19.181s
   CGroup: /system.slice/httpd.service
           └─ 63637 /usr/sbin/httpd -DFOREGROUND
             └─ 292343 /usr/sbin/httpd -DFOREGROUND
               └─ 292344 /usr/sbin/httpd -DFOREGROUND
                 └─ 292346 /usr/sbin/httpd -DFOREGROUND
                   └─ 292372 /usr/sbin/httpd -DFOREGROUND

un 17 08:18:05 ip-172-31-37-91.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
un 17 08:18:05 ip-172-31-37-91.ap-south-1.compute.internal httpd[63637]: Server configured, listening on: port 80
un 17 08:18:05 ip-172-31-37-91.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
un 21 00:00:09 ip-172-31-37-91.ap-south-1.compute.internal systemd[1]: Reloading httpd.service - The Apache HTTP Server...
un 21 00:00:09 ip-172-31-37-91.ap-south-1.compute.internal systemd[1]: Reloaded httpd.service - The Apache HTTP Server.
un 21 00:00:09 ip-172-31-37-91.ap-south-1.compute.internal httpd[63637]: Server configured, listening on: port 80
ec2-user@ip-172-31-37-91 ~]$
```

i-0d1a3305151f767a8 (MyWebServer78)
PublicIPs: 35.154.19.122 PrivateIPs: 172.31.37.91

Step 4: Deploy a Web Page

Edit the default index file:

`sudo echo "<h1>Welcome to My Web Server</h1>" > /var/www/html/index.html`

```
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-06-17 08:18:05 UTC; 3 days ago
     Docs: man:httpd.service(8)
   Main PID: 63637 (httpd)
  Status: "Total requests: 542; Idle/Busy workers 100/0; Requests/sec: 0.00159; Bytes served/sec: 1 B/sec"
    Tasks: 177 (limit: 1014)
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                 └─ 292344 /usr/sbin/httpd -DFOREGROUND
                    └─ 292346 /usr/sbin/httpd -DFOREGROUND
                       └─ 292372 /usr/sbin/httpd -DFOREGROUND

Jun 17 08:18:05 ip-172-31-37-91.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 17 08:18:05 ip-172-31-37-91.ap-south-1.compute.internal httpd[63637]: Server configured, listening on: port 80
Jun 17 08:18:05 ip-172-31-37-91.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jun 21 00:00:09 ip-172-31-37-91.ap-south-1.compute.internal systemd[1]: Reloading httpd.service - The Apache HTTP Server...
Jun 21 00:00:09 ip-172-31-37-91.ap-south-1.compute.internal httpd[63637]: Reloading httpd.service - The Apache HTTP Server.
Jun 21 00:00:09 ip-172-31-37-91.ap-south-1.compute.internal httpd[63637]: Server configured, listening on: port 80
[ec2-user@ip-172-31-37-91 ~]$ sudo ss -tulnp | grep :80
tcp        LISTEN 0      511          *:80          *:.*    users:((("httpd",pid=292372,fd=4), ("httpd",pid=292346,fd=4), ("httpd",pid=292344,fd=4), ("httpd",pid=63637,fd=4))
[ec2-user@ip-172-31-37-91 ~]$ sudo echo "<h1>Welcome to My Web Server</h1>" > /var/www/html/index.html
-bash: /var/www/html/index.html: Permission denied
[ec2-user@ip-172-31-37-91 ~]$ echo "<h1>Welcome to My Web Server</h1>" | sudo tee /var/www/html/index.html
<h1>Welcome to My Web Server</h1>
[ec2-user@ip-172-31-37-91 ~]$
```

i-Od1a3305151f767a8 (MyWebServer78)
PublicIPs: 35.154.19.122 PrivateIPs: 172.31.37.91

Step 5: Access the Web Server

Open a browser and go to:

<http://your-instance-public-ip>

<http://35.154.19.122>



Step 6: Configure Auto Start (Optional)

Ensure the web server starts on reboot:

`sudo systemctl enable httpd`

```
tcpd ,pid=63637,fd=4))
[ec2-user@ip-172-31-37-91 ~]$ sudo echo "<h1>Welcome to My Web Server</h1>" > /var/www/html/index.html
-bash: /var/www/html/index.html: Permission denied
[ec2-user@ip-172-31-37-91 ~]$ echo "<h1>Welcome to My Web Server</h1>" | sudo tee /var/www/html/index.html
<h1>Welcome to My Web Server</h1>
[ec2-user@ip-172-31-37-91 ~]$ sudo systemctl enable httpd
[ec2-user@ip-172-31-37-91 ~]$
[ec2-user@ip-172-31-37-91 ~]$
```

i-Od1a3305151f767a8 (MyWebServer78)
PublicIPs: 35.154.19.122 PrivateIPs: 172.31.37.91

CloudShell Feedback Console Mobile App

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Conclusion-

We have successfully deployed an EC2-based web server running Apache or Nginx. This setup is commonly used for hosting websites, web applications, or acting as a reverse proxy.